



Cambridge International AS & A Level

CHEMISTRY

9701/11

Paper 1 Multiple Choice

June 2026

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.
- Important values, constants and standards are printed in the question paper.

This document has **16** pages.

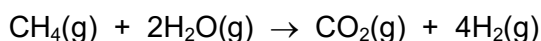
- 1 What is the electronic configuration for the particle ${}_{13}^{27}\text{Al}^{+}$?
- A $1s^2 2s^2 2p^6 3s^1 3p^1$
- B $1s^2 2s^2 2p^6 3s^2$
- C $1s^2 2s^2 2p^6 3s^2 3p^2$
- D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^1$
- 2 The data in the table gives the 5th to the 10th ionisation energies of three elements from Period 3 of the Periodic Table.

element	ionisation energy / kJ mol^{-1}					
	5th	6th	7th	8th	9th	10th
X	6274	21 269	25 398	29 855	35 868	40 960
Y	7012	8 496	27 107	31 671	36 579	43 140
Z	6542	9 362	11 018	33 606	38 601	43 963

What are the correct identities of these three elements?

	element X	element Y	element Z
A	Na	Mg	Al
B	Mg	Al	Si
C	P	S	Cl
D	S	Cl	Ar

- 3 What contains 9.03×10^{23} oxygen atoms?
- A 0.25 mol aluminium oxide
- B 0.75 mol sulfur dioxide
- C 1.5 mol sulfur trioxide
- D 3.0 mol water
- 4 Methane and steam react to produce hydrogen.



0.80 g of methane and 1.35 g of steam react. One of the reactants is used up.

Which volume of hydrogen, measured at room conditions, will be produced?

- A 1.80 dm^3 B 3.60 dm^3 C 4.80 dm^3 D 7.20 dm^3

- 5 Which statement explains why sodium and potassium have different melting points?
- A** The attraction between cations and delocalised electrons is stronger in sodium.
- B** The attraction between cations and anions is stronger in sodium.
- C** The attraction between atoms is stronger in sodium.
- D** The attraction between nuclei and shared electron pairs is stronger in sodium.

- 6 NH_3 and HCN react together to form NH_4CN , an ionic compound.

Which row states the number of coordinate bonds and the number of π bonds in one formula unit of NH_4CN ?

	number of coordinate bonds	number of π bonds
A	0	2
B	1	2
C	0	3
D	1	3

- 7 When 1.0 mol of ethanoic acid, CH_3COOH , in aqueous solution is neutralised by an excess of aqueous sodium hydroxide, 55 kJ mol^{-1} of energy is released.

Which statement about this reaction is correct?

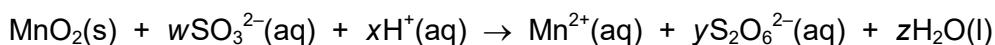
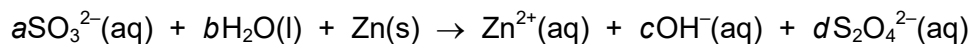
- A** The reaction is exothermic because only bond breaking takes place.
- B** The reaction is exothermic because only bond forming takes place.
- C** The reaction is exothermic because more energy is given out in breaking bonds than is taken in to form bonds.
- D** The reaction is exothermic because more energy is given out in forming bonds than is taken in to break bonds.
- 8 In an experiment, 1.60 g of a fuel is burnt. 45.0% of the energy released is absorbed by 200 g of water. The temperature of the water rises from 18.0°C to 66.0°C .

What is the total energy released per gram of fuel burnt?

- A** 25 100 J **B** 55 700 J **C** 89 200 J **D** 143 000 J

- 9 Sulfite ions, SO_3^{2-} , react separately with zinc and with manganese dioxide.

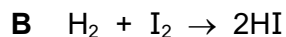
a , b , c , d , w , x , y and z are all whole numbers.



Which numbers are correct for a , b , w and x ?

	a	b	w	x
A	1	2	2	4
B	2	2	2	4
C	1	2	4	2
D	2	2	4	2

- 10 Which equation shows hydrogen acting as an oxidising agent?



- 11 1.00 g of nitrogen gas is stored in a 2.00 dm^3 vessel at 40.0°C .

What is the pressure in the vessel?

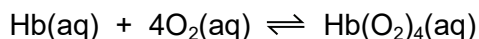
A 5940 Pa

B 11 900 Pa

C 46 400 Pa

D 92 900 Pa

- 12 One molecule of haemoglobin, Hb, can bind with four molecules of oxygen according to the equation shown.



When the equilibrium concentration of O_2 is $7.6 \times 10^{-6} \text{ mol dm}^{-3}$, the equilibrium concentrations of Hb and $\text{Hb}(\text{O}_2)_4$ are equal.

What is the numerical value of K_c for this equilibrium?

A 3.0×10^{20}

B 1.3×10^5

C 7.6×10^{-6}

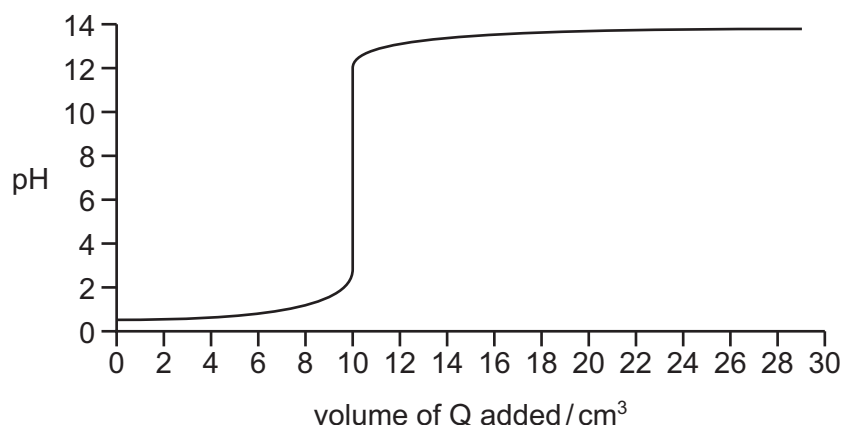
D 3.3×10^{-21}

13 Aqueous acid P and aqueous alkali Q have the same concentration.

20 cm³ of P is added to a conical flask.

Q is slowly added to the flask and the volume of Q and the pH are recorded.

The pH titration curve is shown.

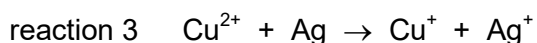
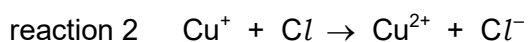
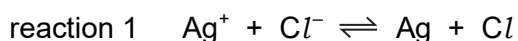


Which row gives the identity of P and Q?

	P	Q
A	HCl	NaOH
B	H ₂ SO ₄	NH ₃
C	HCl	Ba(OH) ₂
D	CH ₃ COOH	Sr(OH) ₂

14 Photochromic glass, used for sunglasses, darkens when exposed to bright light and becomes more transparent again when the light is less bright. The darkness of the glass is due to the presence of silver atoms.

The following reactions are involved.



Which statement about these reactions is correct?

- A** Cu^+ and Cu^{2+} ions act as catalysts.
- B** Cu^+ ions act as an oxidising agent in reaction 2.
- C** Reaction 3 increases the darkness of the glass.
- D** Silver atoms are reduced in reaction 3.

- 15** The reversible reaction between methanol and ethanoic acid liquids is catalysed by adding a small volume of concentrated sulfuric acid.

Two statements about this reaction are listed.

- 1 The sulfuric acid is a homogeneous catalyst.
- 2 The sulfuric acid lowers the activation energy of the reverse reaction.

Which statements are correct?

- A** both 1 and 2
B 1 only
C 2 only
D neither 1 nor 2

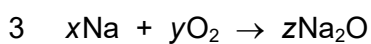
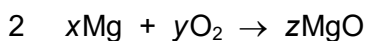
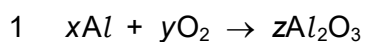
- 16** The atomic radii and ionic radii for three elements in Period 3 are shown.

	atomic radius / nm	ionic radius / nm
element X	0.118	0.053
element Y	0.099	0.180
element Z	0.160	0.072

Using this data, which statement is correct?

- A** Element X has lower electrical conductivity than element Y.
B Element Y has a higher melting point than element X.
C Element Y and element Z react to form an ionic compound.
D Element Z forms ionic compounds by gaining electrons.

- 17** Three equations are listed. x , y and z are all whole numbers.



Which equations can be balanced if $x = 4$ and $z = 2$?

- A** 1 and 3 **B** 1 only **C** 2 only **D** 3 only

- 18** Which oxide has a simple structure rather than a giant structure?

- A** MgO **B** Al_2O_3 **C** SiO_2 **D** P_4O_{10}

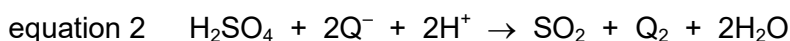
- 19 The trends seen in Group 2 can be used to predict the properties of radium and its compounds.

Which statement is correct?

- A Radium has the highest second ionisation energy of the elements in Group 2.
 - B Radium hydroxide is the least soluble of the hydroxides of the elements in Group 2.
 - C Radium carbonate has the lowest thermal stability of the carbonates of the elements in Group 2.
 - D Radium reacts faster with water than the other elements in Group 2.
- 20 Equal masses of CaCO_3 , $\text{Ca}(\text{NO}_3)_2$, BaCO_3 and $\text{Ba}(\text{NO}_3)_2$ are thermally decomposed. The volume of gas produced in each experiment is measured under the same conditions.

Which compound will produce the greatest volume of gas?

- A CaCO_3 B $\text{Ca}(\text{NO}_3)_2$ C BaCO_3 D $\text{Ba}(\text{NO}_3)_2$
- 21 Equation 1 and equation 2 show two different reactions of halide ion Q^- with concentrated sulfuric acid.



What is Q?

- A Cl^- or Br^- B Br^- or I^- C Cl^- only D I^- only
- 22 What happens when iodine solution is added to a solution of sodium bromide?
- A A reaction occurs without changes in oxidation state.
 - B Bromide ions are oxidised; iodine atoms are reduced.
 - C Bromide ions are reduced; iodine atoms are oxidised.
 - D No reaction occurs.

- 23 Which statement is correct?

- A Nitrogen is unreactive due to the absence of lone pairs in the molecule.
- B Aqueous ammonia contains both $\text{NH}_3(\text{aq})$ and $\text{NH}_4^+(\text{aq})$. The $\text{NH}_4^+(\text{aq})$ ion is a Brønsted–Lowry acid.
- C High temperatures are needed to supply the energy to break the strong double bonds in nitrogen molecules when they react.
- D Atmospheric SO_2 reacts with unburnt hydrocarbons to form a component of photochemical smog.

- 24** When dry ammonia and hydrogen chloride gases are mixed, a solid white ionic compound is formed.

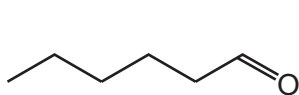
Two statements are listed.

- 1 The formation of the ionic compound is a redox reaction.
- 2 During the formation of the ionic compound, the H–N–H bond angle increases.

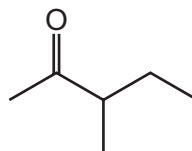
Which statements are correct?

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B 1 only
C 2 only
D neither 1 nor 2

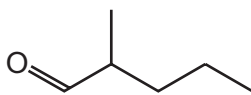
- 25** The diagrams show skeletal formulas of some isomers of $C_6H_{12}O$.



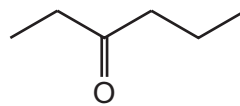
1



2



3

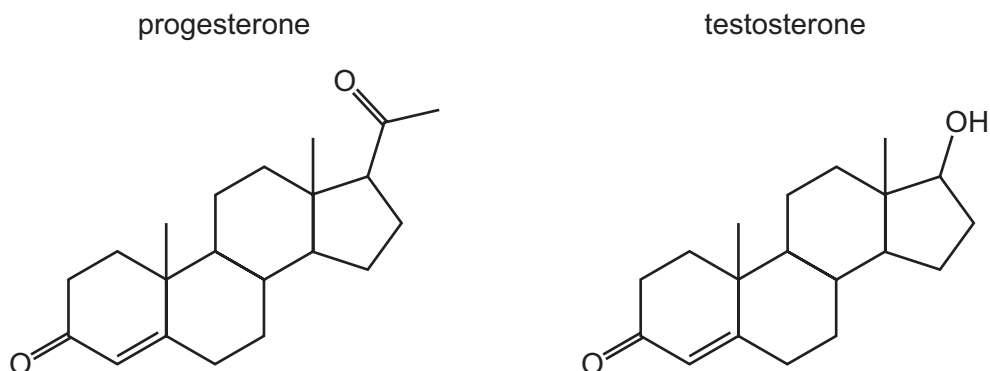


4

Which statement is correct?

- A** 1 and 3 are chain isomers of each other and positional isomers of each other.
B 2 and 3 are functional group isomers of each other and both have a chiral centre.
C 1 and 4 are functional group isomers of each other and both have a chiral centre.
D 2 and 4 are positional isomers of each other and functional group isomers of each other.

26 The skeletal formulas of two compounds are shown.



Which statements about progesterone and testosterone are correct?

- A** They both contain a ketone group, and they both have geometrical isomers due to the C=C bond.
- B** They have the same molecular formula, and they both have geometrical isomers.
- C** They have the same number of chiral carbons, and they have the same molecular formula.
- D** They have the same number of chiral carbons, and they both contain a ketone group.
- 27 Which intermediate ion forms in the greatest amount during the addition of HBr to propene?
- A** $\text{CH}_3\text{CH}^+\text{CH}_3$
- B** $\text{CH}_3\text{CH}_2\text{CH}_2^+$
- C** $\text{CH}_3\text{CH}^-\text{CH}_2\text{Br}$
- D** $\text{CH}_3\text{CHBrCH}_2^-$
- 28 Two hydrocarbons, $\text{R}-\text{CH}_3$ and $\text{R}'-\text{CH}_3$, react separately with bromine in the presence of ultraviolet light. One of these hydrocarbons is unsaturated.

In each case, free radical substitution reactions occur.

R is $\text{CH}_3\text{CHC}(\text{CH}_3)$.

R' is $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2$.

Which row is correct?

	a propagation stage for the saturated hydrocarbon	a termination stage for the unsaturated hydrocarbon
A	$\text{R}-\text{CH}_2\cdot + \text{Br}\cdot \rightarrow \text{R}-\text{CH}_2\text{Br}$	$\text{R}'-\text{CH}_2\cdot + \text{Br}\cdot \rightarrow \text{R}'-\text{CH}_2\text{Br}$
B	$\text{R}-\text{CH}_2\cdot + \text{Br}_2 \rightarrow \text{R}-\text{CH}_2\text{Br} + \text{Br}\cdot$	$\text{R}'-\text{CH}_2\cdot + \text{Br}_2 \rightarrow \text{R}'-\text{CH}_2\text{Br} + \text{Br}\cdot$
C	$\text{R}'-\text{CH}_3 + \text{Br}\cdot \rightarrow \text{R}'-\text{CH}_2\cdot + \text{HBr}$	$2\text{R}-\text{CH}_2\cdot \rightarrow \text{R}-\text{CH}_2\text{CH}_2-\text{R}$
D	$2\text{R}'-\text{CH}_2\cdot \rightarrow \text{R}'-\text{CH}_2\text{CH}_2-\text{R}'$	$\text{R}-\text{CH}_3 + \text{Br}\cdot \rightarrow \text{R}-\text{CH}_2\cdot + \text{HBr}$

29 Which statement about p orbitals is correct?

- A All p orbitals are  shaped.
- B All p orbitals have a lower energy than an s orbital of the same principal quantum number.
- C Each of the first four quantum shells contain p orbitals.
- D Each p orbital can contain a maximum of six electrons.

30 The first four ionisation energies for phosphorus, in kJ mol^{-1} , are shown.

1st	2nd	3rd	4th
1060	1900	2920	4960

Which row gives the 5th to 14th ionisation energies for phosphorus?

	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th
A	6270	8410	21 270	25 400	29 800	35 870	40 100	46 300	59 000	271 800
B	6270	21 270	25 400	29 860	35 870	40 100	46 300	54 070	59 000	271 800
C	6270	21 270	25 400	29 860	35 870	40 100	46 300	59 000	271 800	301 400
D	21 270	25 400	29 860	35 870	40 100	46 300	54 070	59 000	271 800	301 400

31 Sodium carbonate neutralises phosphoric acid to produce sodium phosphate, carbon dioxide and water only.

Which volume of 0.50 mol dm^{-3} sodium carbonate is needed to exactly neutralise 40.0 cm^3 of 0.25 mol dm^{-3} phosphoric acid?

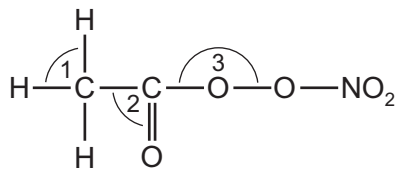
- A 10.0 cm^3 B 13.3 cm^3 C 20.0 cm^3 D 30.0 cm^3

32 Which equation describes the complete combustion of an alkene, C_nH_{2n} ?

- A $\text{C}_n\text{H}_{2n} + 2n\text{O}_2 \rightarrow n\text{CO}_2 + n\text{H}_2\text{O}$
- B $\text{C}_n\text{H}_{2n} + 2n\text{O}_2 \rightarrow n\text{CO}_2 + 2n\text{H}_2\text{O}$
- C $\text{C}_n\text{H}_{2n} + 1.5n\text{O}_2 \rightarrow n\text{CO}_2 + 2n\text{H}_2\text{O}$
- D $\text{C}_n\text{H}_{2n} + 1.5n\text{O}_2 \rightarrow n\text{CO}_2 + n\text{H}_2\text{O}$

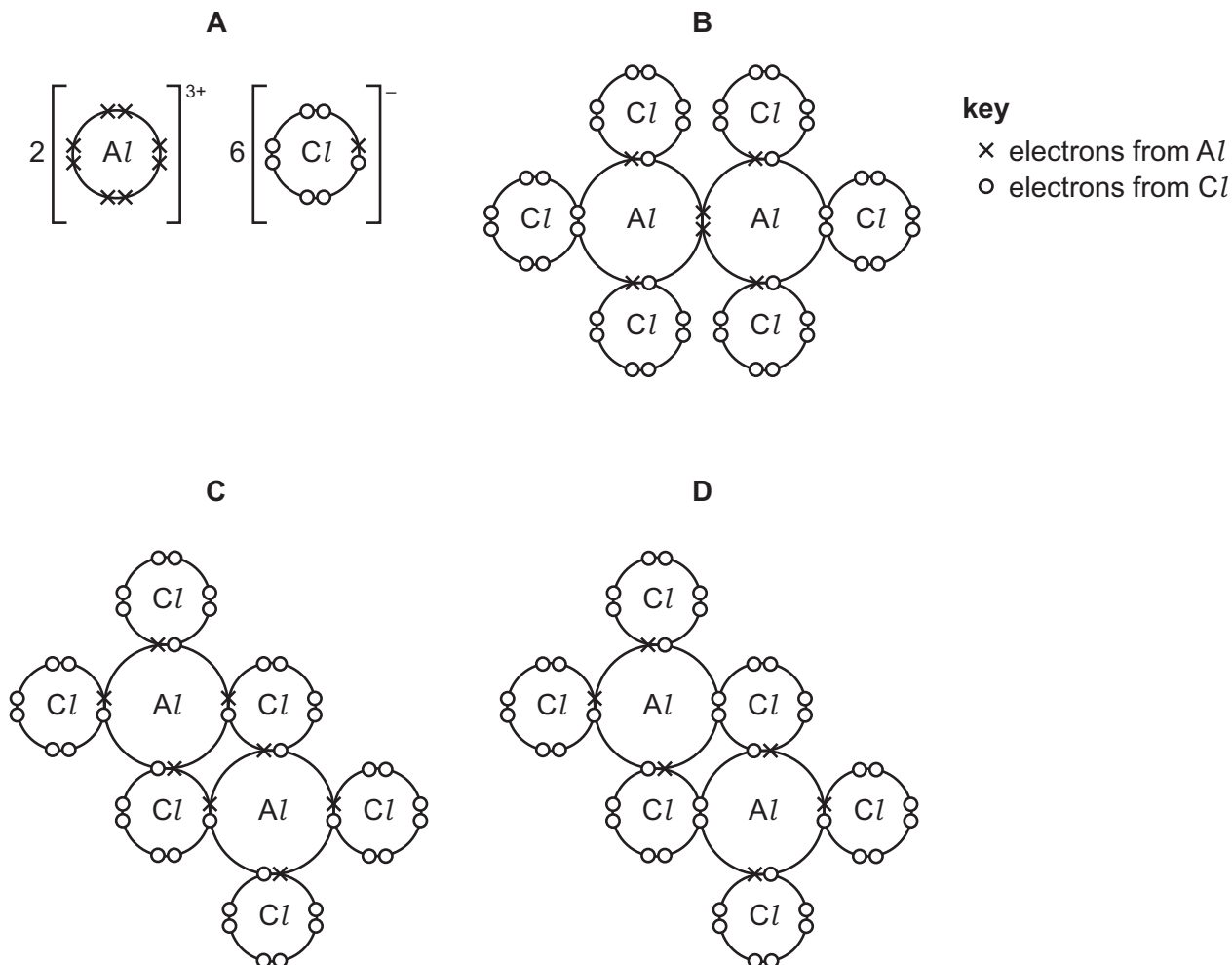
- 33 Organic nitrates in photochemical smog can cause breathing difficulties.

The diagram shows an example of an organic nitrate molecule.

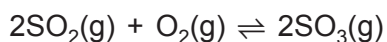


What is the correct order of the bond angles shown in ascending order (smallest first)?

- A** $1 \rightarrow 2 \rightarrow 3$ **B** $2 \rightarrow 1 \rightarrow 3$ **C** $3 \rightarrow 1 \rightarrow 2$ **D** $3 \rightarrow 2 \rightarrow 1$
- 34 Which dot-and-cross diagram is correct for Al_2Cl_6 ?



- 35 The Contact process involves an equilibrium reaction.



Which row gives the K_p expression and the pressure conditions used for this reaction?

	K_p expression	pressure / atm
A	$\frac{p^2\text{SO}_3}{p^2\text{SO}_2 \times p\text{O}_2}$	2
B	$\frac{p^2\text{SO}_2 \times p\text{O}_2}{p^2\text{SO}_3}$	2
C	$\frac{p^2\text{SO}_3}{p^2\text{SO}_2 \times p\text{O}_2}$	200
D	$\frac{p^2\text{SO}_2 \times p\text{O}_2}{p^2\text{SO}_3}$	200

- 36 When nitrogen monoxide gas is mixed with oxygen gas, the molecules collide with each other. Some of these collisions are effective and lead to a reaction.

The pressure of the mixture was increased at constant temperature.

Which row correctly describes how the frequency of effective and non-effective collisions changed?

	frequency of effective collisions	frequency of non-effective collisions
A	increased	decreased
B	increased	increased
C	increased	unchanged
D	unchanged	decreased

- 37 A reversible chemical reaction is studied.

The forward reaction has an activation energy $E_a = 50 \text{ kJ mol}^{-1}$ and an enthalpy change of reaction $\Delta H_f = +20 \text{ kJ mol}^{-1}$.

When a catalyst is added, the activation energy of the forward reaction decreases by 50%.

What is the activation energy of the backwards reaction when the catalyst is present?

- A** 45 kJ mol^{-1} **B** 25 kJ mol^{-1} **C** 15 kJ mol^{-1} **D** 5 kJ mol^{-1}

38 Which substance does **not** give HCl(g) when water is added to it?

- A** MgCl_2 **B** Al_2Cl_6 **C** SiCl_4 **D** PCl_5

39 For Groups 1 and 2, metallic bonding is stronger in metals that have a smaller atomic radius.

X and Y are both metals.

ΔH_{at} is the energy required to produce 1.0 mol of gaseous atoms from the solid.

$$\Delta H_{\text{at}}\text{X} = +107 \text{ kJ mol}^{-1}; \Delta H_{\text{at}}\text{Y} = +148 \text{ kJ mol}^{-1}$$

What are X and Y?

	X	Y
A	Na	K
B	Na	Mg
C	Mg	Na
D	Mg	K

40 The elements in Period 3 and their compounds show trends across the period from sodium to sulfur.

Which row is correct?

	electronegativity of the elements	acid–base behaviour of the oxides of the elements
A	decreases	basic → amphoteric → acidic
B	decreases	acidic → amphoteric → basic
C	increases	basic → amphoteric → acidic
D	increases	acidic → amphoteric → basic

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Important values, constants and standards

molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Faraday constant	$F = 9.65 \times 10^4 \text{ C mol}^{-1}$
Avogadro constant	$L = 6.022 \times 10^{23} \text{ mol}^{-1}$
electronic charge	$e = -1.60 \times 10^{-19} \text{ C}$
molar volume of gas	$V_m = 22.4 \text{ dm}^3 \text{ mol}^{-1}$ at s.t.p. (101 kPa and 273 K) $V_m = 24.0 \text{ dm}^3 \text{ mol}^{-1}$ at room conditions
ionic product of water	$K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ (at 298 K (25 °C))
specific heat capacity of water	$c = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$ (4.18 J g ⁻¹ K ⁻¹)

The Periodic Table of Elements

Group																											
1	2	1										13	14	15	16	17	18										
		Key atomic number atomic symbol name relative atomic mass																									
3	4																	5	6	7	8	9	10	11	12		
Li lithium 6.9	Be beryllium 9.0																	B boron 10.8	C carbon 12.0	N nitrogen 14.0	O oxygen 16.0	F fluorine 19.0					
11	12																	13	14	15	16	17	18				
Na sodium 23.0	Mg magnesium 24.3																	Al aluminium 27.0	Si silicon 28.1	P phosphorus 31.0	S sulfur 32.1	Cl chlorine 35.5	Ar argon 39.9				
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36										
K potassium 39.1	Ca calcium 40.1	Sc scandium 45.0	Ti titanium 47.9	V vanadium 50.9	Cr chromium 52.0	Mn manganese 54.9	Fe iron 55.8	Co cobalt 58.9	Ni nickel 58.7	Cu copper 63.5	Zn zinc 65.4	Ga gallium 69.7	Ge germanium 72.6	As arsenic 74.9	Se selenium 79.0	Br bromine 79.9	Kr krypton 83.8										
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54										
Rb rubidium 85.5	Sr strontium 87.6	Y yttrium 88.9	Zr zirconium 91.2	Nb niobium 92.9	Mo molybdenum 95.9	Tc technetium —	Ru ruthenium 101.1	Rh rhodium 102.9	Pd palladium 106.4	Ag silver 107.9	Cd cadmium 112.4	In indium 114.8	Sn tin 118.7	Sb antimony 121.8	Te tellurium 127.6	I iodine 126.9	Xe xenon 131.3										
55	56	57–71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86										
Cs caesium 132.9	Ba barium 137.3	lanthanoids		Ta tantalum 180.9	W tungsten 183.8	Re rhenium 186.2	Os osmium 190.2	Ir iridium 192.2	Pt platinum 195.1	Au gold 197.0	Hg mercury 200.6	Tl thallium 204.4	Pb lead 207.2	Bi bismuth 209.0	Po polonium —	At astatine —	Rn radon —										
87	88	89–103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118										
Fr francium —	Ra radium —	actinoids		Rf rutherfordium —	Sg seaborgium —	Bh bohrium —	Hs hassium —	Mt meitnerium —	Ds darmstadtium —	Rg roentgenium —	Cn copernicium —	Nh nihonium —	Fl flerovium —	Mc moscovium —	Lv livermorium —	Ts tennessine —	Og oganeson —										